

Abstract

Spatio-temporal data are plentiful and easily obtained. Examples are satellite images of parts of the earth, temperature readings for a number of nearby stations, election results for voting districts and a number of consecutive elections, trajectories for people or animals possibly with additional sensor readings, disease outbreaks or volcano eruptions.

Most active Data mining research areas with respect to space and time can be grouped into three categories: 1) Temporal Association Rule Mining, which aims to detect correlations in transactional and relational data that possess a time component, 2) Spatial Clustering, which aims to group similar objects into the same cluster while grouping dissimilar objects into different clusters. In this case similarity is influenced by both spatial and aspatial attributes of the objects as well as any obstacles that may exist, and 3) Time-series Analysis, which aims to detect frequent patterns in the values of an attribute over time.

Data Mining research with respect to space and time happens conditionally, meaning that either first the spatial aspect is analyzed, after which the temporal aspects are analyzed, or reversed, but not in a joint, integral modeling approach, where space and time are not separated.

Though the spatio-temporal data are often relatively abundant in either space, or time, but not in both. Which is true in the case of data mining as well, significantly, most of the data mining work deals with one or the other of spatial or temporal semantics, with very few handling both.

This paper presents our study and experience in conducting research on spatio-temporal pattern mining on the Air Quality Index of Alabama from 1980 to 2019, built air quality patterns, and present our analysis. By identifying areas with unusual high (hotspot) or low (cold-spot) measures of air quality, the state health administrators can allocate more healthcare resources to those areas that desire more attention.

Introduction

Air pollution is a major environmental pollution which is directly affecting most of the people in the world in some way. Some of them include respiratory and cardiovascular diseases, Global warming, wildfires, etc. Ambient air pollution is a major environmental health problem affecting everyone in low-, middle- and high-income countries and was estimated to cause 4.2 million premature deaths worldwide per year in 2016 (WHO, 2018).

Globally, many countries assess their air quality using various index. In the United States air quality is assessed by Air Quality Index (AQI) which reports daily air quality which was implemented by the United States Environmental Protection Agency (EPA) in 1999. The AQI considers five major health pollutants: Ground-level ozone (O₃), Sulfur dioxide (SO₂), Carbon monoxide (CO), nitrogen dioxide (NO₂) and Particulate matter (PM₁₀ and PM_{2.5}). The AQI has a range from 0 to 500 and has breakpoints (0, 50, 100, 150, 200, 300, 500) which determines the condition of the air and was designed to be understandable to the public.

The effect of single pollutants on human health is calculated as

$$I_p = \frac{(I_{HI} - I_{LO})}{BP_{HI} - BP_{LO}} (C_p - BP_{LO}) + I_{LO}$$

where
I_p=Index for pollutant P
C_p=Rounded concentration of pollutant P
BP_{HI}=Break point that is greater than or equal to CP
BP_{LO}=Breakpoint that is less than or equal to CP
I_{HI}=AQI value corresponding to BPHI
I_{LO}=AQI value corresponding to BPLO

The highest individual pollutant index, I_p, represents the Air Quality Index (AQI) of the location (Kanchan et al., 2015).

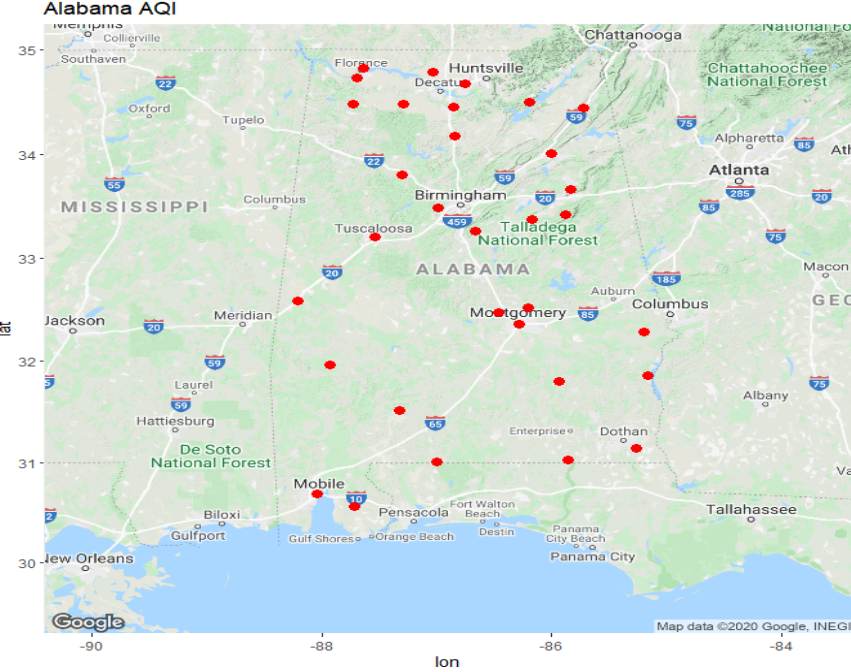
Data mining is the process of finding anomalies, patterns and correlations within large data sets to predict outcomes. Using various techniques, we can use this information to find insights and discover knowledge from the data. Thus, the knowledge can be applied to further research in the topic.

Spatio-temporal data mining is the process of discovering patterns and knowledge from spatiotemporal data. Spatiotemporal data is the data which involves both space and time.

The data contains 21 rows and 721 columns. The column contains state, county, different levels of healthy and unhealthy days, different levels of AQI, days with different chemicals like CO, NO₂, Ozone, SO₂, PM₂, PM₁₀, and the latitude and longitude of each county.

Data Analysis

We used R programming language which is software environment for statistical computing and graphics.

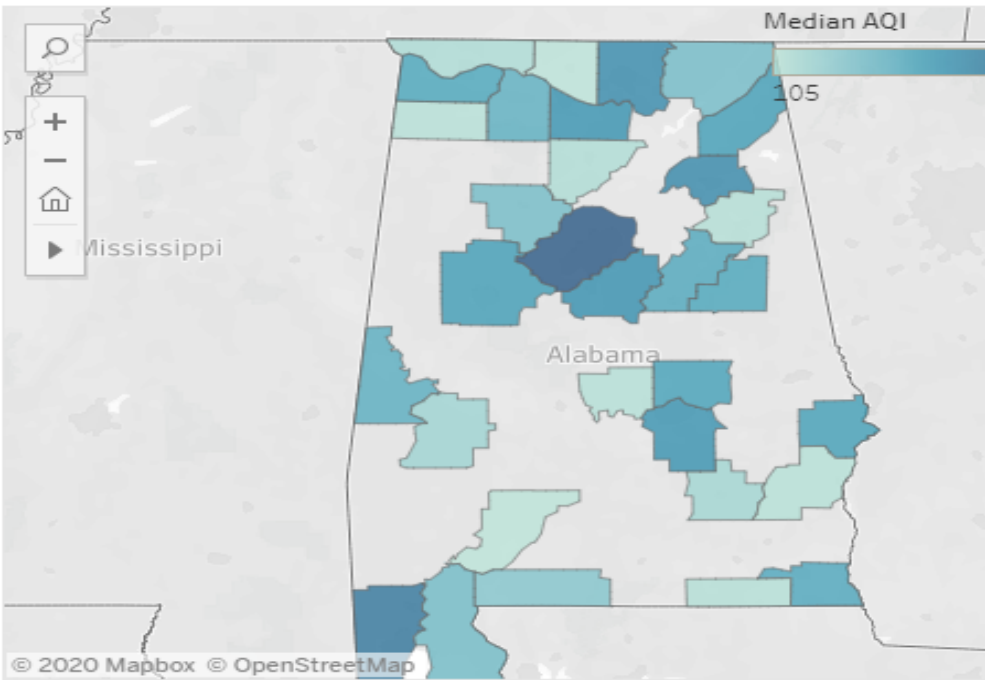
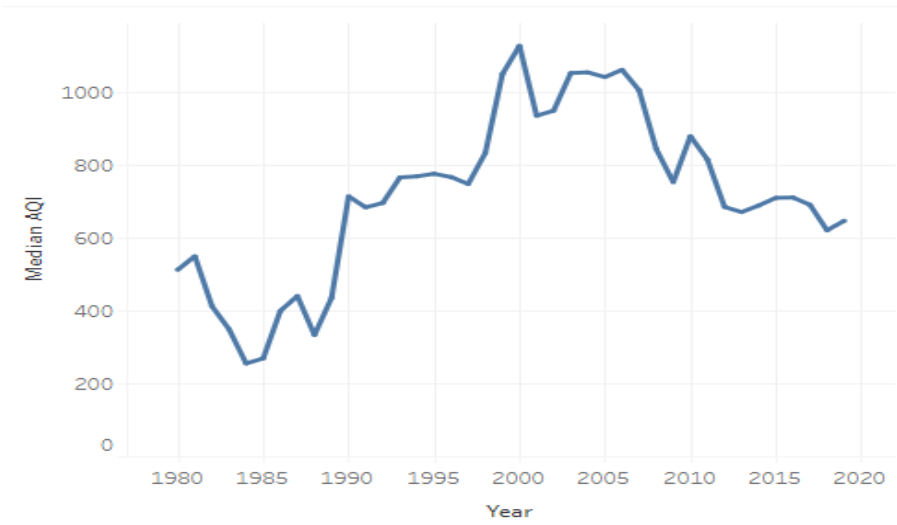


The above red spots on the map are the counties of Alabama that have been plotted in the graph using the latitude and longitude parameters in the dataset where the AQI is analyzed. The plot below shows the increase

of Median AQI over the years which is a

AQI Value	Air Condition	Color Symbol
0-50	Good	Green
51-100	Moderate	Yellow
101-150	Unhealthy for sensitive group	Orange
151-200	Unhealthy	Red
201-300	Very Unhealthy	Purple
301-500	Hazardous	Maroon

concern for the authorities. Moreover, we used the k-means clustering method to find the low and hot spot. The next figure shows that Jefferson County has the highest median AQI in Alabama with Mobile county having the second highest Median AQI. Monroe and Limestone county had one of the lowest Median AQI in Alabama.



Conclusion

The paper presents our study and experience in conducting research on spatio-temporal pattern on the Air Quality Index of Alabama from 1980 to 2019. We have identified areas with unusual high and unusual low measures of Air quality.

Improvements

We can further use advanced statistical computing software to analyze and gain deep insights about the data.

References

- Kanchan, Amit Kumar Gorai and Pramila Goyal (2015) A Review on Air Quality Indexing System Asian Journal of Atmospheric Environment Vol. 9-2, pp. 101-113 doi:http://dx.doi.org/10.5572/ajae.2015.9.2.101
- WHO (2018) Ambient(outdoor) air pollution

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